ALGEBRA 1 · UNIT 4: RELATIONS AND FUNCTIONS

Roller Coaster Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students analyze the Thunder Ridge roller coaster: they decide which relations are functions, read domain, range, and key features from a height-versus-time graph, use function notation, model the line with an arithmetic sequence, and design and graph a coaster of their own.

One familiar context ties together the whole unit: relations and the vertical line test (a loop!), domain and range, function notation, key features of graphs, discrete vs. continuous, and arithmetic sequences. The design task makes students build a graph with given features.

Standards Addressed

- F-IF.1 Understand functions, domain, and range
- F-IF.2 Use function notation and evaluate functions
- F-IF.4 Interpret key features of graphs
- F-IF.5 Relate the domain to its context
- F-IF.3 and F-BF.2 Arithmetic sequences as functions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: play a coaster video and sketch height vs. time together. Part 1: relations and functions.
Day 2	Part 2: key features and function notation on the Thunder Ridge graph.
Day 3	Part 3: the queue as an arithmetic sequence, then design the coaster.

Materials

- Student handout
- Rulers and colored pencils
- Graph paper for the coaster design
- Optional: video of a coaster ride (to match with the graph)

How the Three Versions Differ

Level	What changes
CP2	Reads the graph with guided prompts, mapping-diagram function check, and an arithmetic sequence with a table.
CP1	Adds a formula for the lift hill, average rate of change, discrete vs. continuous domains, and explicit and recursive sequence rules.
Honors	Writes the graph as a piecewise function, solves h(t) = 45 on every piece, and designs a coaster with constraints on rates of change.

Watch For (Common Misconceptions)

- Swapping domain and range.
- Thinking a loop-shaped track passes the vertical line test because the ride is real.
- Reading h(20) as 'h times 20'.

Extension Ideas

- Have students record a video of a playground slide and graph it.
- Add speed vs. time and compare the two graphs.
- Connect slopes of the pieces to Unit 5 (rate of change).

Answer Key – CP2

Part 1: Is It a Function?

#	Task	Answer
1	Each rider is assigned to a car. Is "rider → car" a function? Explain.	Yes. Each rider is assigned to exactly one car.
2	Now reverse it: "car → rider." Is this a function? Explain.	No. Car 1 is matched with two riders (Ana and Ben).
3	Which of these relations are functions? A = {(1, 4), (2, 6), (3, 8), (4, 10)}; B = {(5, 2), (5, 7), (6, 3)}; C = {(0, 3), (1, 3), (2, 3)}	A and C are functions. B is not: the input 5 has two outputs.
4	Use the vertical line test. Which track graph is a function? Why can't the other one be a function?	Track B is a function. Track A has a loop, so a vertical line crosses it more than once.

Part 2: Read the Ride

#	Task	Answer
5	What is the domain of the ride? What is the range?	Domain $0 \le t \le 41$; range $10 \le h \le 90$
6	Find h(0), h(20), and h(30). Explain what h(20) means.	h(0) = 10, h(20) = 90, h(30) = 60. At 20 seconds the car is 90 feet high.
7	When is the car at its highest point? At which times is the car at its lowest height?	Highest at t = 20 s. Lowest (10 ft) at t = 0, t = 25, and t = 41 s.
8	List the time intervals when the car is going up.	0 to 20 s, 25 to 30 s, and 33 to 36 s
9	Which drop is the steepest? How do you know from the graph?	The drop from 20 to 25 s (90 ft down in 5 s); the graph is steepest there.

Part 3: The Line & Your Coaster

#	Task	Answer
10	Complete the table.	5, 6.5, 8 for groups 3, 4, 5
11	Write a rule for the wait time w(n) of group n. How long does group 20 wait?	$w(n) = 2 + 1.5(n - 1) = 1.5n + 0.5$; w(20) = 30.5 minutes
12	Design your own coaster! Sketch a height vs. time graph that lasts 30 to 60 seconds, has at least two hills, and has a top height between 100 and…	Answers vary.

Answer Key – CP1

Part 1: Is It a Function?

#	Task	Answer
1	Is "rider → car" a function? Is "car → rider"?	Rider → car is a function; car → rider is not (car 1 has two riders).
2	Which relation is a function? A = {(1, 4), (2, 6), (3, 8)}; B = {(5, 2), (5, 7), (6, 3)}; C = {(0, 3), (1, 3), (2, 3)}. Give the domain and range…	A: function, domain {1, 2, 3}, range {4, 6, 8}. B: not a function. C: function, domain {0, 1, 2}, range {3}.
3	Explain, using the vertical line test, why Track A is not a function of horizontal position but every ride's height IS a function of time.	A vertical line through the loop hits Track A twice. In time, the car can only be at one height at each moment.
4	Ride tickets cost $3 each and a wristband limit is 8 rides. Is the cost a discrete or continuous function of rides? Give its domain and range.	Discrete. Domain {0, 1, 2, …, 8}; range {0, 3, 6, …, 24} dollars.

Part 2: Read the Ride

#	Task	Answer
5	Give the domain and range. Identify the maximum and every local maximum.	Domain $0 \le t \le 41$, range $10 \le h \le 90$. Maximum 90 at $t = 20$; local maxima 60 at $t = 30$ and 45 at $t = 36$.
6	The lift hill is $h(t) = 4t + 10$ for $0 \le t \le 20$. Find $h(12)$ and solve $h(t) = 60$. Interpret both.	$h(12) = 58$ ft at 12 s. $4t + 10 = 60 \rightarrow t = 12.5$ s.
7	Find the average rate of change from $t = 20$ to $t = 25$ and from $t = 0$ to $t = 25$. Explain why the second one is 0 even though the car moved.	$(10 - 90) \div (25 - 20) = -16$ ft per s. $(10 - 10) \div 25 = 0$: the car ends at the same height it started.
8	Write the increasing and decreasing intervals.	Increasing: $0 < t < 20$, $25 < t < 30$, $33 < t < 36$. Decreasing: $20 < t < 25$, $30 < t < 33$, $36 < t < 41$.

Part 3: The Line & Your Coaster

#	Task	Answer
9	Write explicit and recursive rules for the wait $w(n)$. Is the domain discrete or continuous?	$w(n) = 1.5n + 0.5$; $w(1) = 2$, $w(n) = w(n - 1) + 1.5$. Discrete (whole-number groups).
10	The park posts a sign when the wait is more than 45 minutes. Which group is the first to see the sign?	$1.5n + 0.5 > 45 \rightarrow n > 29.67 \rightarrow$ group 30
11	Design a coaster (30 to 60 s, at least two hills, top height 100 to 150 ft). Draw it, give the domain and range, all extrema, intervals, and the…	Answers vary.

Answer Key – Honors

Part 1: Functions & Relations

#	Task	Answer
1	Explain why "rider → car" is a function but its inverse relation is not. What would have to be true for the inverse to be a function?	Each rider gets one car. The inverse is not a function because car 1 has two riders. Each car would need exactly one rider.
2	For which values of k is {(1, 4), (2, k), (k, 8)} a function?	If k = 1, input 1 has outputs 4 and 8. If k = 2, input 2 has outputs 2 and 8. Any other value of k makes it a function.
3	Explain, with the vertical line test, why the loop track is not a function of horizontal position, and describe a different input that would make…	A vertical line through the loop crosses the track twice. Using time as the input makes it a function.
4	Wristband: 8 rides for $20, or $3 per ride without a band. Write a rule for the cheapest cost C(r) for r rides (0 ≤ r ≤ 8) and give its range.	C(r) = 3r for r ≤ 6 and C(r) = 20 for r = 7 or 8 (since 3 · 7 = 21 > 20). Range {0, 3, 6, 9, 12, 15, 18, 20}.

Part 2: The Piecewise Ride

#	Task	Answer
5	Write h(t) as a piecewise function with six linear pieces (include each domain).	h(t) = 4t + 10 (0 ≤ t ≤ 20); −16t + 410 (20 < t ≤ 25); 10t − 240 (25 < t ≤ 30); −10t + 360 (30 < t ≤ 33); 5t − 135 (33 < t ≤ 36); −7t + 297 (36 < t ≤ 41)
6	Find every time t when h(t) = 45. Show which pieces you used.	4t + 10 = 45 → t = 8.75; −16t + 410 = 45 → t ≈ 22.81; 10t − 240 = 45 → t = 28.5; −10t + 360 = 45 → t = 31.5; t = 36 (the local max). Five times.
7	Find the average rate of change on each piece. Which piece is the fastest climb? The fastest drop?	Rates: 4, −16, 10, −10, 5, −7 ft per s. Fastest climb 10 (25 to 30 s); fastest drop −16 (20 to 25 s).
8	Park rule: no drop may be faster than 15 ft per second on average. Does Thunder Ridge follow it? If not, propose a new point that fixes it without…	No: the first drop is 16 ft per s. Example fix: move the bottom from (25, 10) to (26, 10): (10 − 90) ÷ 6 ≈ −13.3.

Part 3: Design Within the Rules

#	Task	Answer
9	Design your coaster as a piecewise linear function. List every point and every piece.	Answers vary.
10	Show that every piece follows the rate rules. Give the domain, range, and all extrema.	Answers vary.

11	The queue for your ride grows by 1.5 minutes per group starting at 2 minutes, and your ride loads a new group every 50 seconds. Write the wait as…	w(n) = 1.5n + 0.5. Loading every 50 s means each group adds only 50 s ≈ 0.83 min, so 1.5 min is too much unless groups are larger than the cars.

Scoring note: For the {(1, 4), (2, k), (k, 8)} item: k = 1 gives input 1 two outputs (4 and 8); k = 2 gives input 2 outputs 2 and 8. Any other k works.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Coaster design meets every park rule and its key features are identified correctly. Goes beyond the minimum with a creative, realistic choice.	Coaster design meets every park rule and its key features are identified correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes a labeled graph, domain and range, key features, function notation, and the queue model. Polished, organized, and visually engaging.	Proposal includes a labeled graph, domain and range, key features, function notation, and the queue model. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.